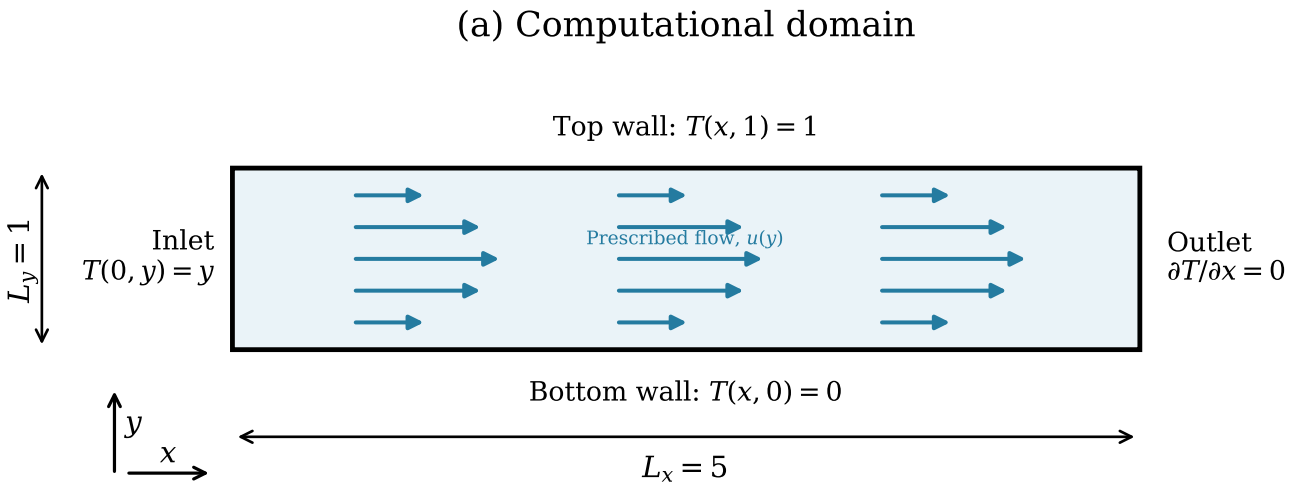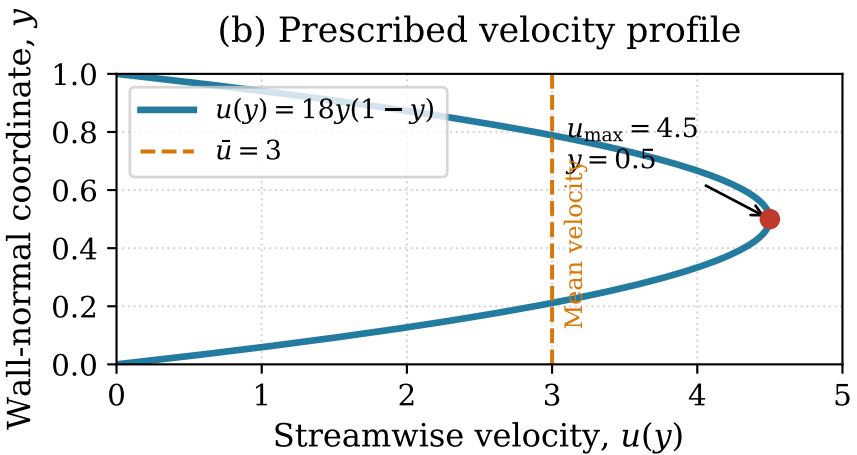Computational domain and prescribed velocity field